Non-linear PDE : Ist order

linear PDE: $px + qy = z$

Non-linear PDE: $px + qy + p^2 + q^2 = z$

Charpit's method for the 1st order non-linear PDE:

The standard form of the non-linear PDE is

$$f(x, y, z, p, q) = 0 \quad \text{——— (1)}$$

We know the total derivative is

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} dy = p\,dx + q\,dy \quad \text{——— (2)}$$

Let another relation be

$$F(x, y, z, p, q) = 0 \quad \text{——— (3)}$$

Charpit's auxiliary equation is

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0}$$

Ex(1). Solve $z = px + qy + p^2 + q^2$ ——— (1)

Ans) Let $f(x, y, z, p, q) = z - px - qy - p^2 - q^2 = 0$

Charpits A.E is

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0} \quad \text{——— (1)}$$

$f_x = -p,\ f_y = -q,\ f_z = 1,\ f_p = -x-2p,\ f_q = -y-2q$

(11) $\Rightarrow \dfrac{dp}{-p+p\cdot 1} = \dfrac{dq}{-q+q\cdot 1} = \dfrac{dz}{p(x+2p)+q(y+2q)} = \dfrac{dx}{x+2p} = \dfrac{dy}{y+2q}$

$dp = 0 \Rightarrow p = a$ (a is a constn

$dq = 0 \Rightarrow q = b$ (b is a const.)

Putting a and b in (1), we get

$z = ax + by + a^2 + b^2$ Ans.

Ex 2: $q = 3p^2$ — (1)

Let $f(x, y, z, p, q) = 3p^2 - q = 0$ — (2)

Char pits A.E is

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\Rightarrow \frac{dp}{0+0} = \frac{dq}{0+0} = \frac{dz}{-p\cdot 6p + q\cdot 1} = \frac{dx}{-6p} = \frac{dy}{1}$$

$\Rightarrow dp = 0 \Rightarrow p = a$

$dq = 0$

(1) $\Rightarrow q = 3a^2$ — (2)

Putting the values of p and q in

$dz = p\,dx + q\,dy = a\,dx + 3a^2\,dy$

$$\int dz = \int a\,dx + \int 3a^2\,dy$$

$$\Rightarrow \boxed{z = ax + 3a^2 y + b}$$

Ex 3: Solve $zpq = p + q$

Ans: $z^2 = 2(1+a)\left[x + \left(\frac{1}{a}\right)y\right] + b$

Ex: Solve $p^2 - y^2 q = y^2 - x^2$ — (1)

Let $f(x, y, z, p, q) = p^2 - y^2 q - y^2 + x^2 = 0$.

Charpit's A.E is

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\Rightarrow \frac{dp}{2x} = \frac{dq}{-2qy - 2y} = \frac{dz}{-p(2p) - q(-y^2)} = \frac{dx}{-2p} = \frac{dy}{y^2} \quad \text{— (2)}$$

$\Rightarrow$ Taking first and fourth fraction

$$\frac{dp}{2x} = \frac{dx}{-2p}$$

$$\Rightarrow -p\,dp = x\,dx$$

$$\Rightarrow -\frac{p^2}{2} - \frac{x^2}{2} = \frac{a^2}{2}$$

$$\Rightarrow \boxed{p^2 + x^2 = a^2} \quad \text{— (3)}$$

$$\Rightarrow p = \sqrt{a^2 - x^2}$$

Solving (1) & (3)

$$(1) \Rightarrow a^2 - y^2 - y^2 q = y^2 - x^2$$

$$-y^2 q = y^2 - a^2$$

$$q = -1 + \frac{a^2}{y^2}$$

$$\boxed{q = a^2 y^{-2} - 1} \quad \text{— (4)}$$

We know,

$$dz = p\,dx + q\,dy$$

$$\int dz = \int \sqrt{a^2 - x^2}\,dx + \int (a^2 y^{-2} - 1)\,dy$$

$$z = \sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + \left(\frac{a^2 y^{-1}}{-1} - y\right) + b$$

$$z = \sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) - a^2 y - y + b$$

Ex: Solve (1) $2zx - px^2 - 2qxy + pq = 0$

Ans: $z = ay + b(x^2 - a)$

(2) $px + qy = pq$ Ans: $az = \frac{1}{2}(y + ax)^2 + b$

(3) $pxy + pq + qy = yz$

Ans:

(4) $pxy + pq + qy = yz$

Ans: $\ln(z - ax) = y - a\ln(ax + y)$

(5) $2(z + xp + yq) = yp^2 + \ln b$

A: $z = \frac{ax}{y^2} + \frac{a^2}{4y^3} + \frac{b}{y}$.

Charpit's method:

(5) $p(q^2+1)+(b-z)q=0$

Ans: $2\sqrt{a(z-y)-1}=x+ay+b$.

(6) $p^2x+q^2y=z$.

(7) $(p^2+q^2)y=qz$.

(8) $z=pq$